CADENCE (2A)
Amazon Content Review Sentiment Analysis

AI Studio Final Project

December 6, 2025

BREAK THROUGH TECH

In 2022, Amazon reported that 125 million customers contributed to over 1.5 billion reviews… yet **90% of companies** admit that they **cannot effectively analyze their customer feedback at scale**

Meet Our Team

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We built an AI system that

identifies product features,

analyzes sentiment,

and generates improvement recommendations

to support data-driven product development.

Business Impact

Keep Your Customers

- This model will enhance retention by helping bring attention to and address pain point before customer churn signals appear.
- Even 1% improvement in $5.2B* revenue base yields $52M annual impact.

Product Development Efficiency

- Reduces R&D waste by prioritizing features with demonstrated customer demand.
- Cadence’s Q1 2025 R&D spend of $439M (35.4% of revenue) could see 5-10% efficiency gains worth $22-44M annually.

Customer Experience Enhancement

- Customer feedback reveals specific areas where the AI system would deliver immediate value:
  - Usability Issues
  - Documentation Gaps
  - Performance Bottlenecks
  - Feature Gaps
  - Integration Pain Points
- Faster overall response to customer needs.

*Cadence reported a trailing twelve months (TTM) revenue of $5.21 billion as of September 30, 2025.

OUR APPROACH

01

- Business Understanding
- Data Preparation

02

- Model Building
- Solve the business problem

03

- Dashboard Development
- Delivering the result
- Final Presentation

DATA PREPRATION

DATASET

Hugging Face

Datasets: McAuley-Lab/**Amazon-Reviews-2023**

- 571.54M User Reviews
- 34 Product Categories
- 48.19M Total Product Items

Examples

★★★★★ 5.0

"Best schematic capture tool I've used in years!"

This design suite is incredible. The interface is intuitive, the simulation runs smoothly, and the integration with layout tools is seamless. I will say it takes some time to master all the features, though.

★☆☆☆☆ 1.0

"Frustrated with constant crashes and poor support."

I really wanted to like this tool but it's been a nightmare. The software crashes during long simulations, losing hours of work. The library management is confusing and documentation is outdated.

DATA PREPROCESSING

DATA COLLECTION & UNDERSTANDING

- Loaded 34 Product Categories.
- Included titles, full review text, star ratings, and metadata.
- Provides broad coverage across different product types, ensuring feature and sentiment patterns are not biased to one category.

CLEANING & NORMALIZING TEXT

- Removed duplicates to prevent repeated reviews from screwing feature-level sentiment.
- Normalized the content review data (e.g. punctuation removal, consistent formatting) to reduce noise in the reviews, allowing the system to focus on actual product features and opinions.

SENTIMENT LABELING & TRAINING PREP

- Mapped rating stars to sentiment labels to help the model learn what positive and negative feedback actually look like.
- 1★ → Negative
- 5★ → Positive
- Filtered out ambiguous mid-range ratings to give the system cleaner examples to learn from.

MODEL BUILDING & TRAINING

Our Model

Sentiment Analysis

Reads and interprets text (tweets, reviews, comments, etc.)

Identifies the sentiment:

Positive (e.g., “I love this product!”)
Negative (e.g., “This was a terrible experience.”)
Neutral (e.g., “It arrived yesterday.”)

Some advanced models also detect:

Emotion (happy, sad, angry, excited)
Intensity (strongly positive, mildly negative, etc.)

Our Model

01

BERT-base CASED

02

Twitter RoBERTa

03

Aspect Based Sentiment Analysis (ABSA)

BERT-base case

What It Means

Preserves the original capitalization of each word.
Treats differently-capitalized words as different tokens.

Why It Matters

Capitalization often changes meaning:
“US” (United States) ≠ “us” (pronoun)
“Apple” (company) ≠ “apple” (fruit)

Pros

Faster and lighter than BERT-Large
Easier to train and less prone to overfitting

Twitter RoBERTa

What It Means

The model is trained on millions of short, informal posts (e.g., tweets, comments, chats).
It becomes highly effective at interpreting short reviews, noisy text, and everyday writing.

Why It Matters

Real-world users rarely write in perfect grammar.
Slang and informal cues often reveal true sentiment.

Pros

Short reviews
Slang or noisy text

Aspect Based Sentiment Analysis ABSA

*to be deployed

Why It Matters

Identifies sentiment toward specific features (aspects)

Example

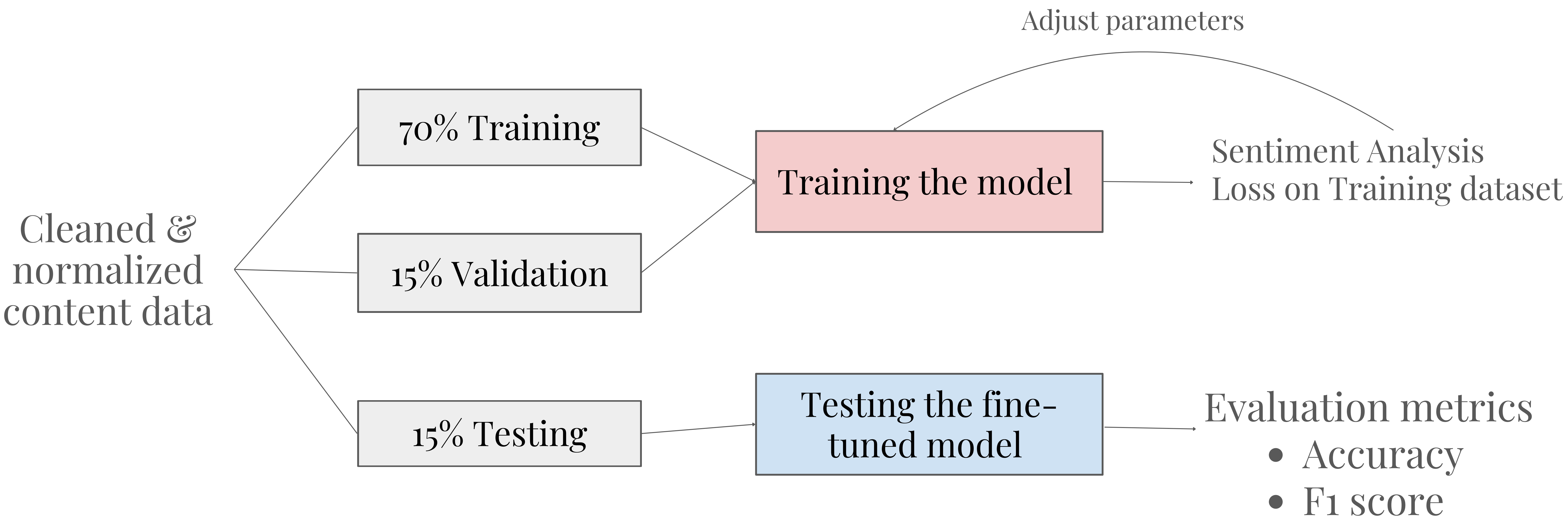
“The battery life is terrible, but the screen is amazing.”

ABSA separates:

Battery life → Negative

Screen → Positive

Model Training & Testing Architecture



RESULTS AND EVALUATION

Fine-Tuning Results

Fine-tuning BERT Model significantly improved model performance across all metrics

87.3%

Accuracy increased from 8% pretrained

86.0%

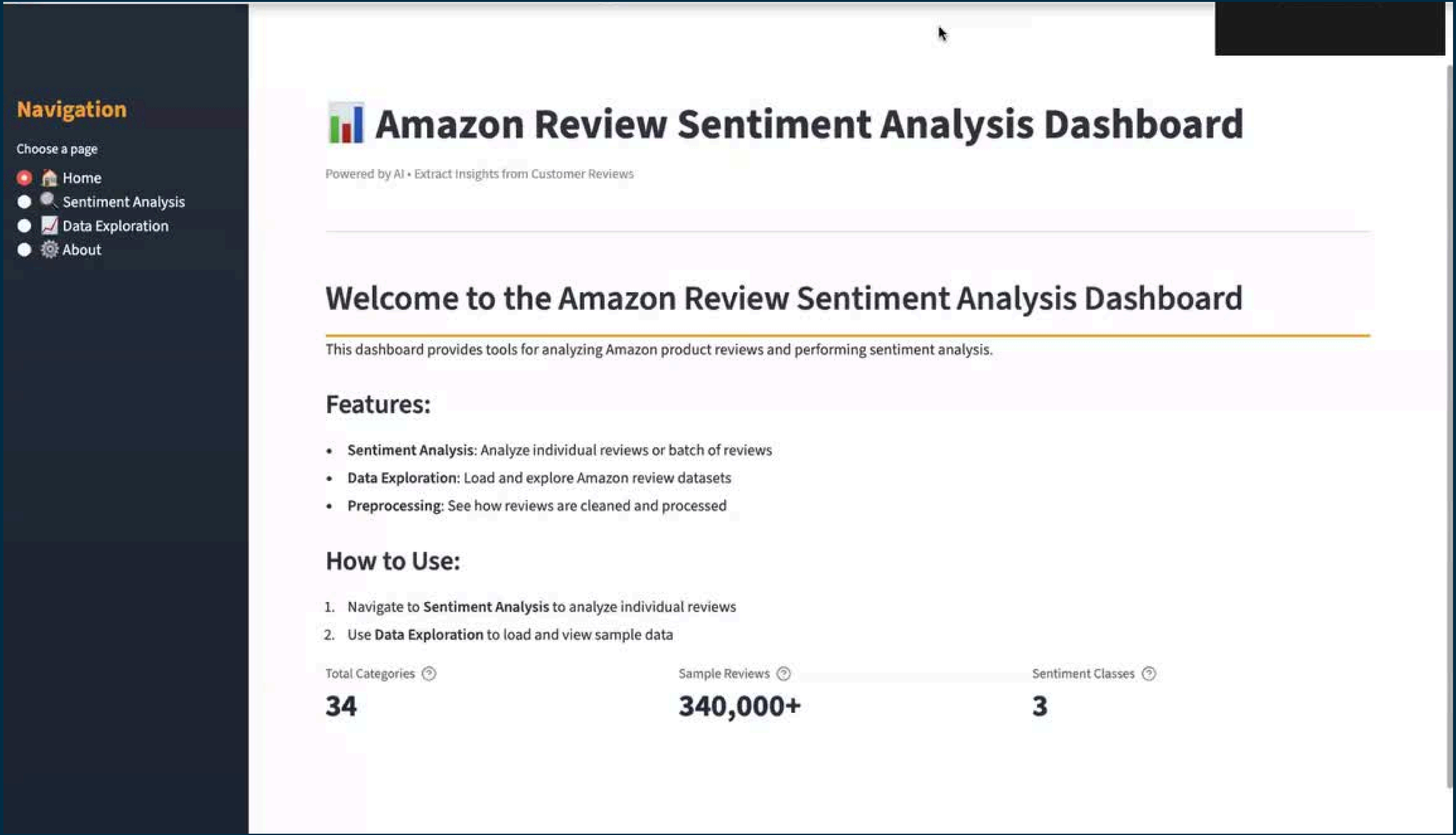
F1 score increased from 1% pretrained

RESULTS AND EVALUATION

- Precision + Recall were both relatively high and balanced
  - Reduces the risk of overlooking critical complaints or over-flagging neutral comments
- Strong F1 Score = consistent feature-level sentiment

Metric	Score
Accuracy	0.873
Precision	0.852
Recall	0.873
F1 Score	0.86

DEMO


Navigation
Choose a page
Home
Sentiment Analysis
Data Exploration
About
Amazon Review Sentiment Analysis Dashboard
Powered by AI • Extract Insights from Customer Reviews
Welcome to the Amazon Review Sentiment Analysis Dashboard
This dashboard provides tools for analyzing Amazon product reviews and performing sentiment analysis.
Features:
Sentiment Analysis: Analyze individual reviews or batch of reviews
Data Exploration: Load and explore Amazon review datasets
Preprocessing: See how reviews are cleaned and processed
How to Use:
1. Navigate to Sentiment Analysis to analyze individual reviews
2. Use Data Exploration to load and view sample data
Total Categories
34
Sample Reviews
340,000+
Sentiment Classes
3

FUTURE DIRECTIONS

INTEGRATE ABSA INTO DASHBOARD

- Add feature-level sentiment visualization
- Highlight most-praised and most-criticized product attributes

IMPROVE OUR FEATURE EXTRACTION PIPELINE

- Move beyond simple keyword detection
- Test a small Named Entity Recognition (NER) model to more reliably identify product features

BUILD A REAL-TIME REVIEW MONITORING PIPELINE

- Look at new Amazon reviews continuously and update streamlit dashboard automatically
- Trigger alerts when negative sentiment spikes for products

THANK YOU

QUESTIONS?